Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/03/2006
PI Name:	Rubin, Clinton Ph.D.		
Project Title:	Biomechanical Countermeasure for Disuse Osteopenia		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Bone Loss Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Stony Brook	**State:**	NY
Zip Code:	11794-2580	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	02/01/2001	**End Date:**	07/31/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hadjiargyrou, Michael (State University of New York)		
Grant/Contract No.:	NCC 9-58-BL.002.08		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Osteoporosis, the progressive loss of bone density and strength which cripples tens of millions on our planet, distinguishes itself as perhaps the greatest physiologic obstacle to an extended human presence in space. Harnessing bone's strong sensitivity to mechanical signals, there is increasing evidence that extremely low magnitude (<100 microstrain) mechanical signals can be strongly osteogenic if applied at a high frequency (15 to 60 Hz). Such high frequency low magnitude strains comprise a dominant component of a bone's strain history, indicating that these mechanical events represent a significant determinant of bone morphology. With this in mind, we have been examining if small perturbations in high frequency loading, induced non-invasively into the lower appendicular skeleton, will stimulate an increase in bone mass without sacrificing bone quality. The principal objectives of this proposal are to establish the efficacy of a unique, biomechanical countermeasure to inhibit bone loss in an animal model of disuse osteoporosis, and correlate this regulatory influence to the expression patterns of several genes critical to bone formation and resorption. The principal objectives of this proposal have been to establish the efficacy of this unique,		

Task Description:	biomechanical countermeasure to inhibit bone loss in an animal model of disuse osteoporosis, and correlate this regulatory influence to the expression patterns of several genes critical to bone formation and resorption. Ten minutes per day of these low level signals (0.25g), induced non-invasively using an oscillating platform, are able to retain bone mass despite 23 hours and 50 minutes of disuse, while ten minutes of normal weightbearing fails to do to. Longer term animal studies (one year), have shown that low level mechanical loading, inducing cortical strains on the order of 5 microstrain, can increase cancellous bone volume fraction, thicken trabeculae, increase trabecular number and enhance bone stiffness and strength. Considering these strain levels are far below (<1/1000th) those which may cause damage to the tissue, we believe these signals hold great potential as a mechanical prophylaxis for osteoporosis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understand how gravity influences the quantity and quality of bone, and how mechanical information may be used to prevent and/or reverse osteoporosis, a disease which effects over 20 million americans here on earth. Ultimately, our goal is to use Low Magnitude Mechanical Stimuli (LMMS) to inhibit or reverse osteoporosis in the human.
Task Progress:	The use of the mouse is facilitating many aspects of the protocol, including comprehensive genomic profiling and expedited access to spaceflight. Considering that many flight opportunities are brief and thus do not permit long term morphologic adaptations in bone to occur, combining the molecular with the tissue level strategies will facilitate establishing countermeasure efficacy even following short term exposure to microgravity. In essence, this work represents a critical step in establishing a physiologically based, non-pharmacologic, non-invasive treatment for osteoporosis, for use on earth or in space. Thus far, the project has demonstrated that the low-level mechanical intervention is powerfully anabolic, and can effectively inhibit the bone loss which parallels disuse. As importantly, using three distinct strains of mice, we have determined that the response of the skeleton to disuse is strongly dependent on the genomic makeup of the animal, and that the responsivity of the skeleton to the signals are dependent on the animal strain and the site examined. These findings point to a unique, non-pharmacologic, non-invasive means of controlling bone loss in a microgravity environment. This biomechanical intervention may potentially displace the need for time consuming (and relatively ineffectual) exercise regimens, or replacing the need for pharmacologic countermeasures (and the potential long-term side effects that they may cause). Importantly, promising results from preliminary clinical trials on post-menopausal women, girls with osteoporosis, and children with cerebral palsy, also indicate that this therapy may work for the 20 million people on earth who suffer from osteoporosis. This work may also contribute to identifying the genetic basis for those at greatest risk of the diseases.
Bibliography Type:	Description: (Last Updated: 10/22/2009)
Articles in Peer-reviewed Journals	Judex S, Garman R, Squire M, Donahue LR, Rubin C. "Genetically based influences on the site-specific regulation of trabecular and cortical bone morphology." J Bone Miner Res. 2004 Apr;19(4):600-6. Epub 2004 Jan 5. PMID: 15005847 , Apr-2004
Articles in Peer-reviewed Journals	Judex S, Garman R, Squire M, Busa B, Donahue LR, Rubin C. "Genetically linked site-specificity of disuse osteoporosis." J Bone Miner Res. 2004 Apr;19(4):607-13. Epub 2004 Jan 5. PMID: 15005848 , Apr-2004
Articles in Peer-reviewed Journals	Squire M, Donahue LR, Rubin C, Judex S. "Genetic variations that regulate bone morphology in the male mouse skeleton do not define its susceptibility to mechanical unloading." Bone. 2004 Dec;35(6):1353-60. PMID: 15589216 , Dec-2004
Articles in Peer-reviewed Journals	Judex S, Zhong N, Squire ME, Ye K, Donahue LR, Hadjiargyrou M, Rubin CT. "Mechanical modulation of molecular signals which regulate anabolic and catabolic activity in bone tissue." J Cell Biochem. 2005 Apr 1;94(5):982-94. PMID: 15597385 , Apr-2005
Articles in Peer-reviewed Journals	Zhong N, Garman RA, Squire ME, Donahue LR, Rubin CT, Hadjiargyrou M, Judex S. "Gene expression patterns in bone after 4 days of hind-limb unloading in two inbred strains of mice." Aviat Space Environ Med. 2005 Jun;76(6):530-5. PMID: 15945395 , Jun-2005